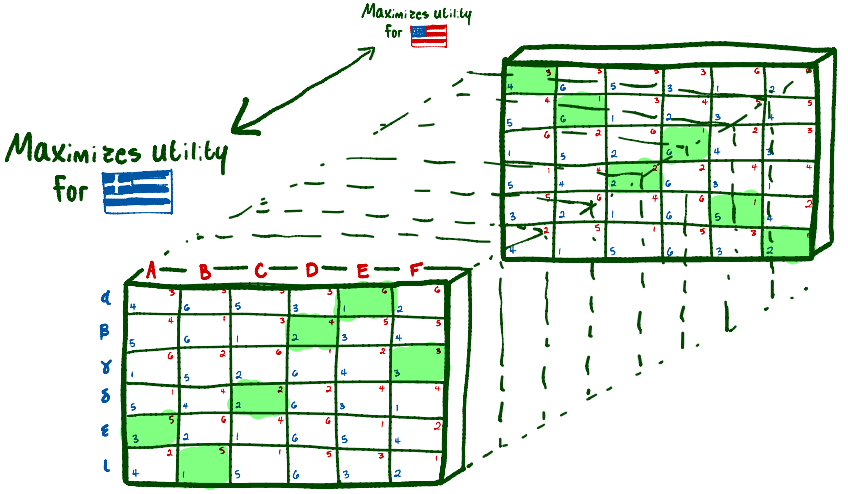
College Admissions and the Stability of Marriage



A presentation of

David Gale & Lloyd Shapley's 1962 paper

by Spencer Elkington

CRITICAL QUESTION

At the end of a wonderful first date, I slide my date a crisp $20 bill from across the table to pay them for their time.

Determine probability p that we have a second date.

There are certain contexts where money should not be used to determine allocations of goods due to moral, contextual, or practical limitations.

Smart market design is key to solve allocation issues in these repugnant markets.[1]

NO INVISIBLE HANDS

ALLOWED

- School Admissions
- Organ Donations
- Child Adoptions
- Political Influence

and, of course,

MARRIAGE

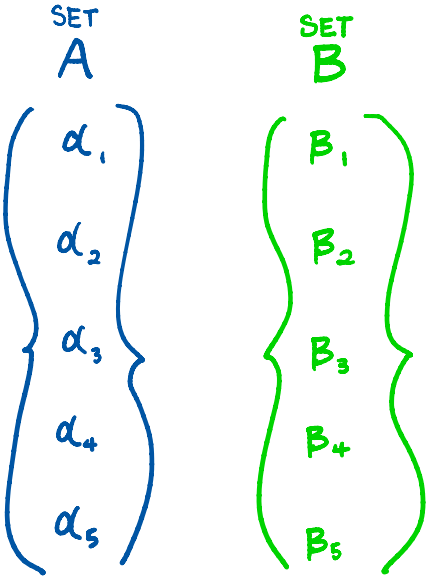
[1] This is the way Alvin Roth, who shares the 2012 Nobel Prize in Economics with Lloyd Shapley, described these illiquid markets in his book "Who Gets What, And Why?"

A Stable marriage problem exists in any market where:

- There exists two disjoint sets of players;
- Every player has a strict preference list of players of the opposite set;
- Players must be paired off such that no two players prefer each other over their assigned partners.

(This final requirement is what makes a matching configuration stable!)

LET'S MAKE A MATCH!



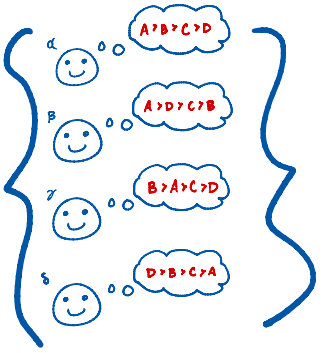
In their 1962 paper,
D. Gale and L. S. Shapley
formalized the stable marriage
problem (SMP) and devised the

Gale-Shapley Stable Matching Algorithm

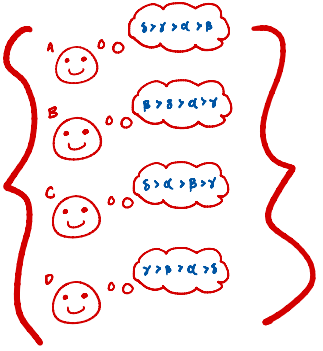
to find stable match
configurations for any SMP!

CASE: PAIRING EXCHANGE STUDENTS

THE GREEKS



THE AMERICANS



RANKING MATRIX

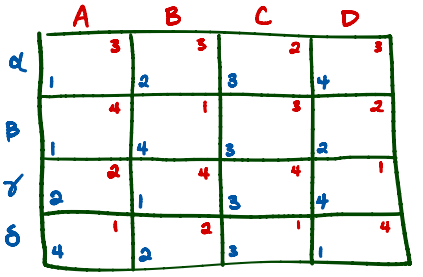
	A	B	C	D
α	3 1	3 2	2 3	3 4
β	4 1	1 4	3 3	2 2
γ	2 2	4 1	4 3	1 4
δ	1 4	2 2	1 3	4 1

COMPUTE:

Gale-Shapley(GREEKS, AMERICANS)

STAGE ONE

- Each Greek proposes to their favorite American.
- Every American accepts the proposal from only their favorite Greek they received a proposal from, but remains willing to renege if a preferred proposal comes along.



α ⇒ A; ✓

β ⇒ A; ✗ A declines because α > β in A's preferences.

γ ⇒ B; ✓

δ ⇒ D; ✓

Lonely Greeks:

β

PROPOSALS

A [∅]
B [∅]
C [∅]
D [∅]

⇓

A [α]
B [γ]
C [∅]
D [δ]

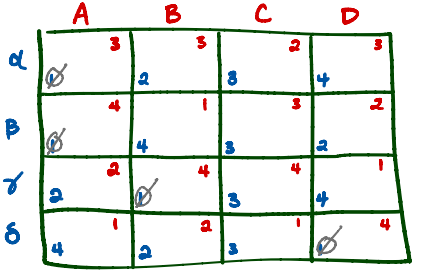
COMPUTE:

Gale-Shapley(GREEKS, AMERICANS)

STAGE 2-N

- While there are still lonely Greeks:
  - Lonely Greeks will propose to their most-preferred American that they have not yet proposed to.
  - If an American receives a proposal from a Greek they prefer over their current proposal, they will renege their previous proposal in favor of the new proposal.

Once the list of lonely Greeks is exhausted, everybody locks in their proposals.



$\beta \Rightarrow D$, ✓ D reneges because $\beta > \delta$ in D's preferences

PROPOSALS

A	α
B	γ
C	∅
D	δ

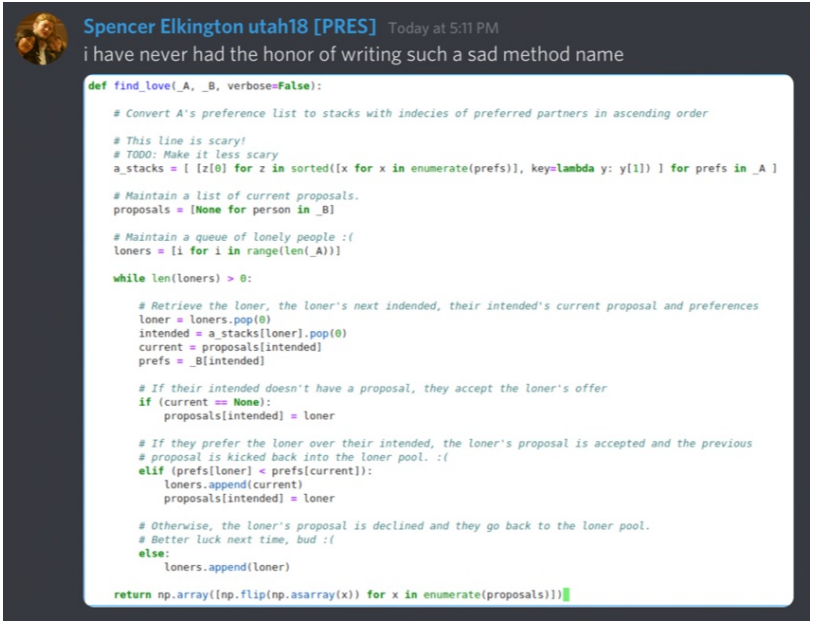
Lonely Greeks:

δ

A	α
B	γ
C	∅
D	β

COMPUTE:

Gale-Shapley (GREEKS, AMERICANS)



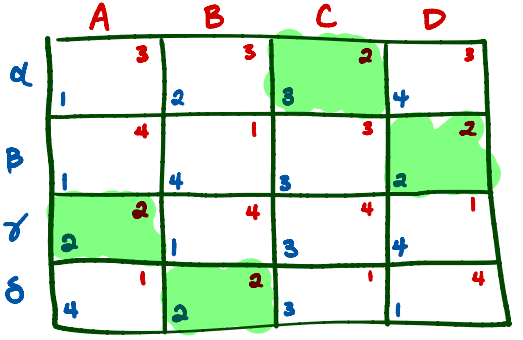
STAGE 1 — α ⇒ A; ✓ — β ⇒ A; ✗ — γ ⇒ B; ✓ — δ ⇒ D; ✓ — A B C D — [α γ ∅ δ] β

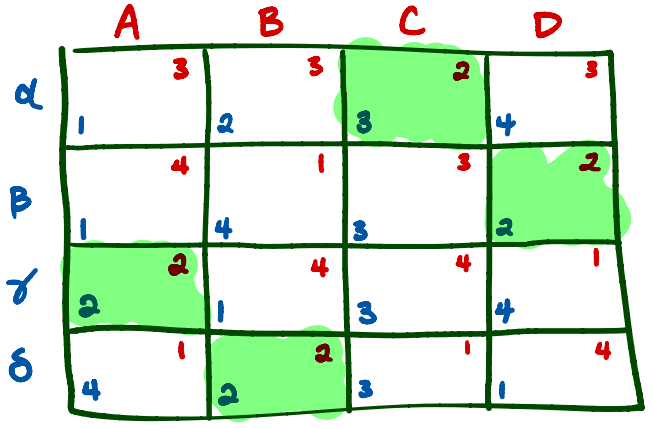
STAGE 2 — β ⇒ D; ✓ R! — [α γ ∅ β] δ

STAGE 3 — δ ⇒ B; ✓ R! — [α δ ∅ β] γ

STAGE 4 — γ ⇒ A; ✓ R! — [γ δ ∅ β] α

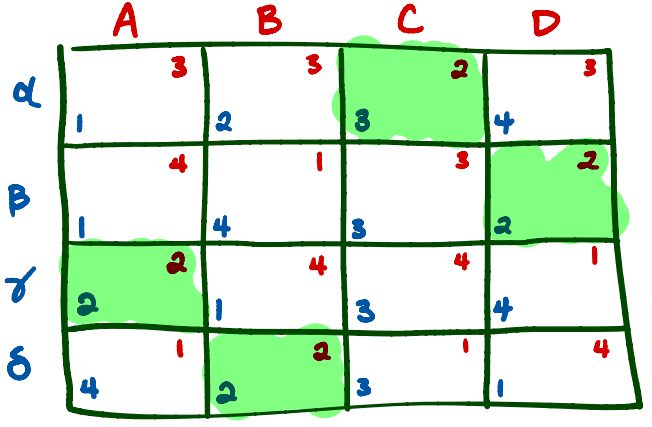
STAGE 5 — α ⇒ C; ✓ — [γ δ α β] ∅





Properties of Stability

- Nobody in the solution set is able to profitably renege on their current partner
- If there exists no solution such that the players could coordinate their own redistribution, it is an optimal stable matching.



1. Proof of Pair Stability

2. Proof of Coverage

3. Proof of The Existence of Stable Configurations

1 Proof of Pair Stability

Any (A, α) pair within a G.S. match set is stable

Suppose there is some pair A and α who prefer each other over their G.S.-assigned partners.

If this were true, α would have proposed to A before proposing to their current partner.

When A received this proposal, they would have reneged on their current partner for α OR not accepted a proposal from their current partner over α.

In either case, A and α would not have been matched by G.S. with their current partners over each other.

Therefore, any pairing produced by G.S. must be stable.

(2) Proof of Coverage

G.S. will produce a stable match set for any equal-sized pair of preference sets

Let A be the proposing preference set of size n, and:

Let B be the accepting preference set, also of size n.

The G.S. process completes once all n members of the proposal set have formed an engagement.

Each engagement requires one element from set A and one element from set B to be formed, and no element from either set may be in more than one engagement.

It follows, then, that if all n members of set A are engaged, then all members of set B are also engaged, and that all members put into G.S. have been matched.

Since all members have been paired and by (1) all pairs are stable, it follows that G.S. creates a stable match configuration for sets A and B.

③

Proof of The Existence of Stable Configurations

There exists at least one stable configuration for all equal-sized pairs of preference sets.

Let A and B be any sets of size n elements that are strict transitive preferences, where the preferences in set A are of elements of set B and vice versa.

From ②, G.S. can find a stable match configuration for A and B.

Therefore, because G.S. can find a stable configuration for any A and B, there must exist at least one stable configuration for any A and B.

Does Order Matter?

INPUT

	A	B	C	D	E	F
α	3 4	3 6	5 5	3 3	6 1	6 2
β	4 5	1 6	3 1	4 2	5 3	5 4
γ	6 1	2 5	6 2	1 6	2 4	3 3
δ	1 5	4 4	2 2	2 6	4 3	4 1
ε	5 3	6 2	4 1	6 6	1 5	2 4
ι	2 4	5 1	1 5	5 6	3 3	1 2

Stable Match A

	A	B	C	D	E	F
α	3 4	3 6	5 5	3 3	6 1	6 2
β	4 5	1 6	3 1	4 2	5 3	5 4
γ	6 1	2 5	6 2	1 6	2 4	3 3
δ	1 5	4 4	2 2	2 6	4 3	4 1
ε	5 3	6 2	4 1	6 6	1 5	2 4
ι	2 4	5 1	1 5	5 6	3 3	1 2

Stable Match B

	A	B	C	D	E	F
α	3 4	3 6	5 5	3 3	6 1	6 2
β	4 5	1 6	3 1	4 2	5 3	5 4
γ	6 1	2 5	6 2	1 6	2 4	3 3
δ	1 5	4 4	2 2	2 6	4 3	4 1
ε	5 3	6 2	4 1	6 6	1 5	2 4
ι	2 4	5 1	1 5	5 6	3 3	1 2

Gale-Shapley is a NON-COMMUNATIVE Function

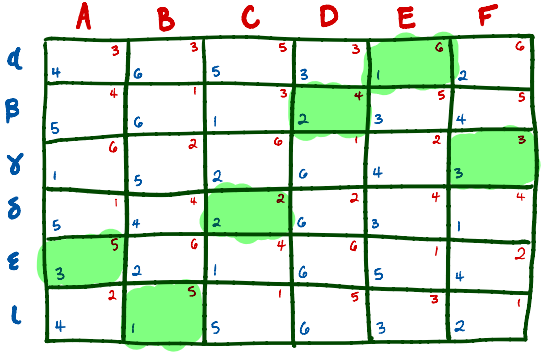
In the same way that

$a \div b \neq b \div a$,

$GS(A, B) \neq GS(B, A)$

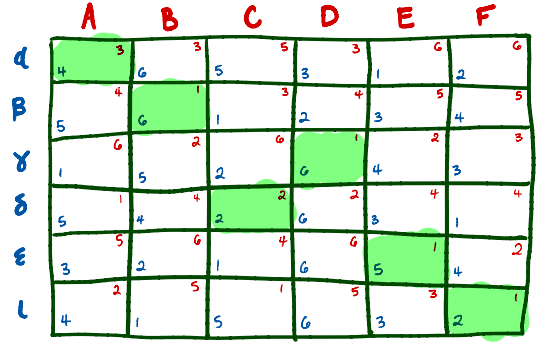
SET-OPTIMAL SOLUTIONS

GS(🇬🇷 , 🇺🇸)



Maximizes utility for 🇬🇷

GS(🇺🇸 , 🇬🇷)



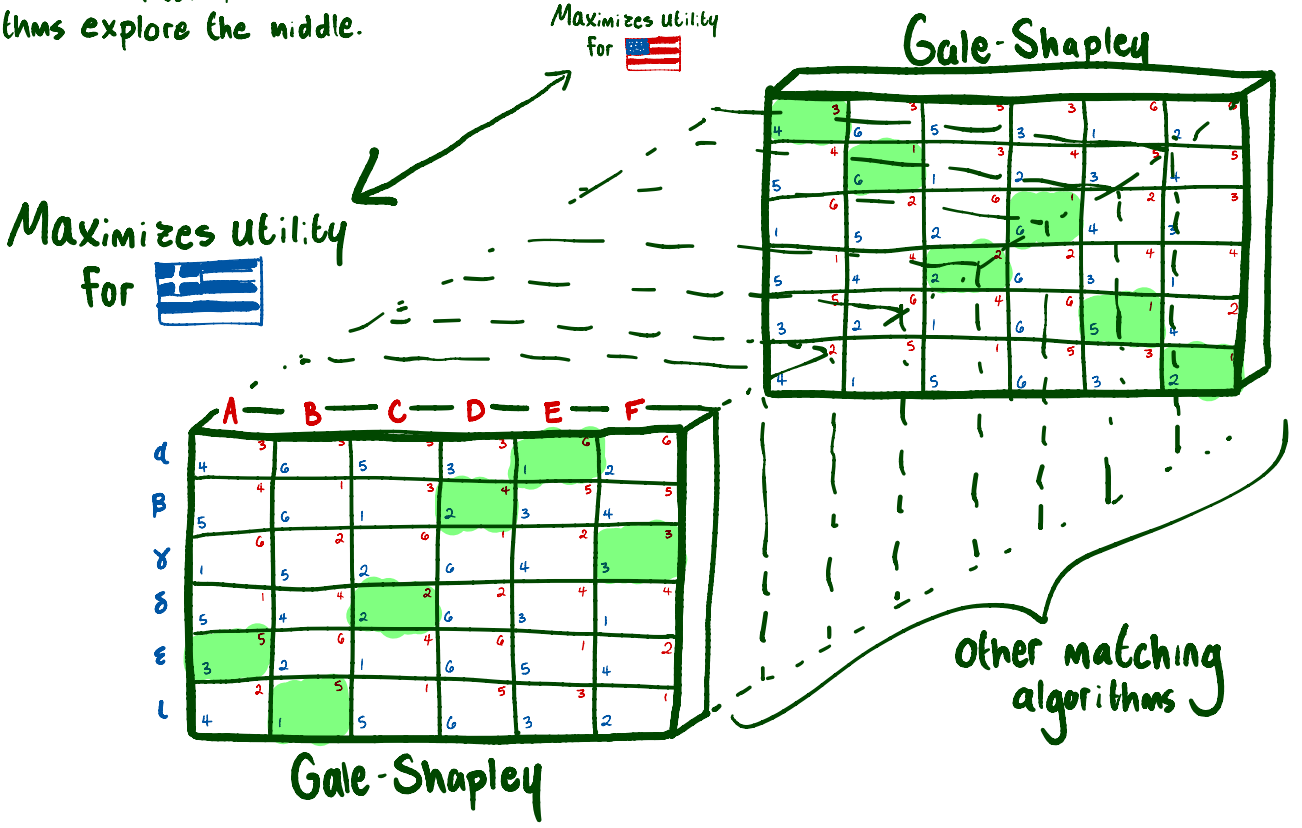
Maximizes utility for 🇺🇸

When designing a market, it is important to consider the practical and ethical implications of the algorithms you use.

(In the original paper that used men and women, the demonstration put men in the proposing position, resulting in a male-optimal solution)

THE STABLE MATCH LATTICE

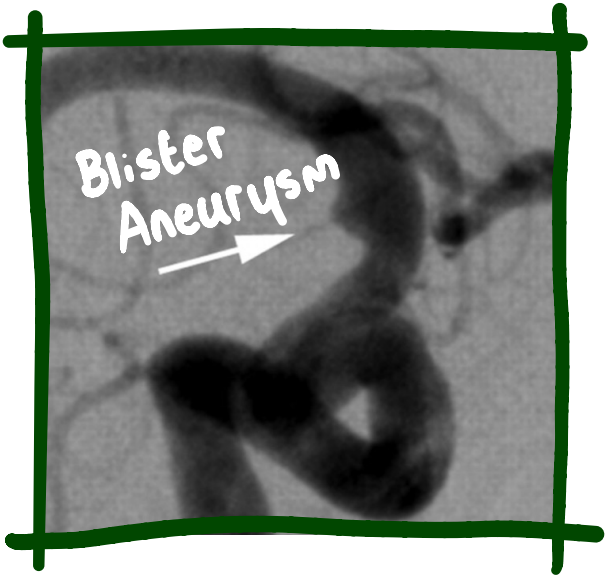
Gale-Shapley computes the ends of the lattice, while other algorithms explore the middle.



Why does this matter?

SEPTEMBER
15
TO-DO:
STABLE MARRIAGE

THIS PAPER will BLOW your MIND

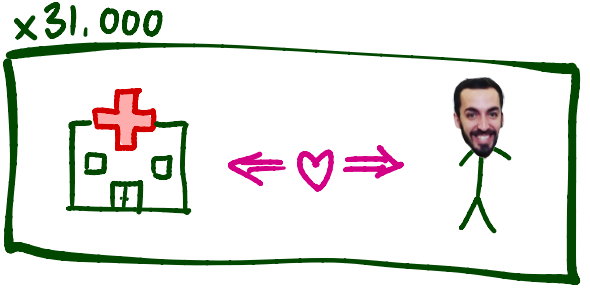


M	T	W	Th	F	Sat	Sun
14	15	16	17	18	19	20
BRAIN SURGERY	STABLE MARRIAGE	BE ALIVE (TENTATIVE)	BE ALIVE (TENTATIVE)	TGIF	TALK LIKE A PIRATE DAY	Q-QUIZ

AUSTIN

Personal Friend
&
Neurosurgeon-in-Training

RESIDENCIES



LAISSEZ FAIRE

Time-Sensitive
Exploding Offers

VS.

CLEARING HOUSE

Non-strategic
Stable Pairing
Systems

FIN